

practices. There are rising concerns regarding privacy and physical safety of children. There are claims that smart toys record conversations, which are translated and converted to text, and personal information is shared with third parties.

Risks are high with toys that are equipped with such unsecured devices as cameras, speakers, sensors, voice/face recognition features, etc. A hacker can easily take a photograph or video of children and listen in on conversations without the kids or parents having the slightest knowledge about it. Poorly-protected toys even allow the hacker to initiate a conversation with children, and children being unable to distinguish the realistic voice coming from the toy from that of the hacker end up disclosing information the hacker intends to extract.

Some smart toys also come with GPS features, as in fitness trackers and smartphones. These are quite vulnerable to revealing user location that increases the threat of kidnapping, predatory stalking and so on.

In addition, the issue of any unauthorised Bluetooth devices connecting to the toy(s) is yet to be solved. Bluetooth normally requires other devices to be relatively close to share information. Here, a security issue arises, which allows any device in Bluetooth range to connect to the toys/apps, which makes it possible for malicious entities to easily snoop on the activities and conversations of children.

Security precautions

Security flaws in connected smart toys have sparked concerns that could allow strangers to steal kids' personal information. Here are a few precautions one can take to help protect the privacy and physical safety of children.

- Check the privacy policy of the company on how data collected from the toy is stored and how it will be processed or used. It is significant to do research before purchasing to know whether the company has a clean reputation with regards to security.

- Determine what kind of data is stored, whether voice messages, facial recognition, Internet-use history, etc. It is crucial to find out if data is encrypted, as encrypted data will hamper the

hacker's attempt of exploiting the data collected.

- Avoid toys that do not require authentication, such as PIN or passwords, when pairing with Bluetooth or Wi-Fi. Hackers can gain easy, unauthorised access to data by attacking unprotected Bluetooth-connected toys that do not need passwords.

- Check if information/data collected from the toy is stored internally or on the cloud. If data is stored on the device, then it is safe. When it is transmitted to external sources, there are security risks with involvement of third parties.

- Last, it is important to figure out if the toy company issues notification on latest software and security updates, or whether software is updated automatically. Ideally, updates should be done on time to protect devices.

Market trends

Despite vulnerabilities due to security, the global market has seen significant growth of connected toys. As per reports, the global smart toys market is expected to reach US\$ 24.65 billion by 2025, growing at a CAGR of 15.5 per cent from 2017 to 2025. Positive trends in the market are attributed to increasing awareness of how technology-led education plays a crucial role in the overall development of the child. In fact, smart toys are being considered as the most effective medium for developing the mental, physical and intellectual skills of children.

Other favourable factors driving the market are increasing use of handheld devices like smartphones and tablets, easy accessibility of the Internet, growth of app-enabled smart toys and so on.

Talking about trends in the Indian market, popularity of smart toys is growing, which indicates a considerable positive prospect for the market. However, what restrains full-fledged growth in the market is the high cost of smart toys, security issues and a threat of cyber-attacks. Interestingly, prominent players in the market are aiming to expand their presence more vigorously through online platforms that will also help to enhance their visibility and gain more consumers. **EFY**

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